Absolute error

1 State the absolute error of the following:

a A measurement of $30\ \text{kg}$, which was measured to the nearest kilogram.

b A measurement of $400\ \text{g}$, which was measured to the nearest gram.

c The amount $\$700$, which was calculated to the nearest $\$10$.

2 A chef measures the amount of olive oil to be used in a cup marked in millilitres. Find the absolute error of any measurement she makes.

3 State the absolute error of each of the following measurements obtained from measuring devices:

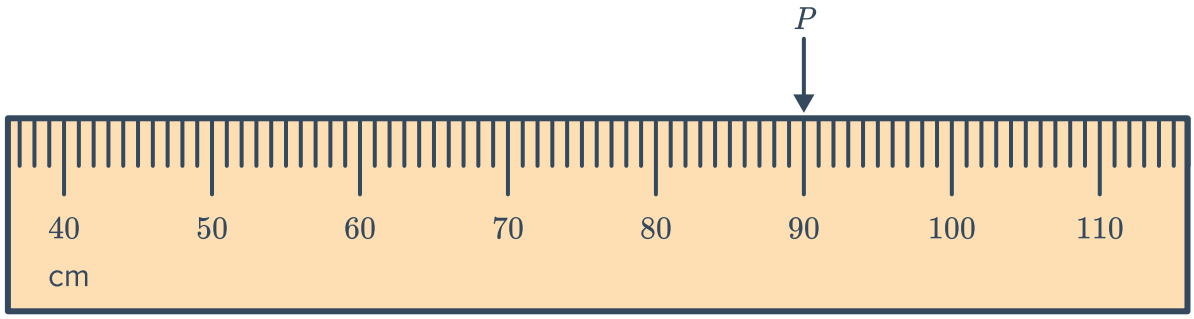
a $18.9\ \text{mL}$ **b** $269\ \text{mm}$ **c** $9.9\ \text{g}$ **d** $18\ \text{s}$

e $15.12\ \text{kg}$ **f** $17.8\ \text{m}$ **g** 117.38°C

4 The number of people at a concert is approximately 5050. Explain why we cannot find the absolute error of this approximation.

The limits of accuracy

5 Consider the tape measure below:



a What is the smallest unit marked on the tape?

b What should be recorded for measurement P?

c What is the absolute error of measurement P?

d State whether the following could be the actual measurement of P:

i $89.02\ \text{cm}$ **ii** $90.64\ \text{cm}$ **iii** $89\ \text{cm}$ **iv** $90.03\ \text{cm}$

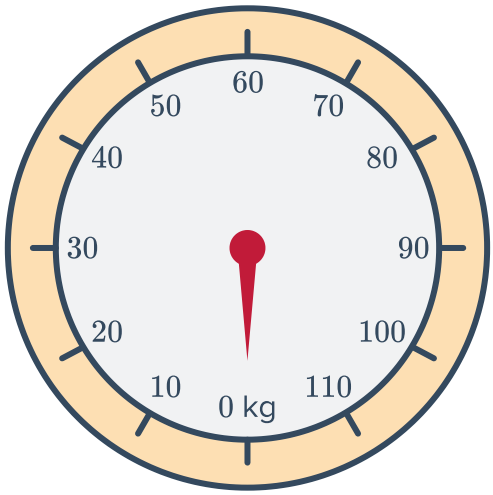
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6 Consider the following scale:

a What is the smallest unit labelled on the scale?

b What is the absolute error?

c If a certain object is measured at $70\ \text{kg}$, what are the lower and upper bounds of this measurement?



7 Consider the speedometer below:

a At what speed is being indicated on the speedometer?

b What is the absolute error of this speed reading?

c What would the maximum speed limit need to be to ensure the driver is not speeding?



8 For the following measurements, find:

i The upper bound **ii** The lower bound

a A distance measured to be $13.45\ \text{km}$.

b A distance measured to be $6.4\ \text{km}$.

c A height measured to be $5\ \text{m}$.

d A bag of sugar weighs $14\ \text{kg}$ to the nearest 10 grams.

e Puncak Jaya, Indonesia's highest mountain, is $4884\ \text{m}$ high rounded to the nearest metre.

f The cost of a CD lie if it is known to be $\$50$ correct to the nearest $\$5$.

9 The length of a piece of rope is measured to be $19.99\ \text{m}$ using a ruler. What is the upper bound of the largest possible length of this rope?

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10 The height of a tower is measured as $2.1\ \text{m}$. What is the shortest possible height of the tower?

11 State the maximum and minimum possible number of the following:

- **a** A town's population is estimated to be 170 people, to the nearest 10 people.
- **b** A stack of paper is estimated to have 3000 sheets to the nearest 100 sheets.

Further calculations (Extended)

12 $c = 4$ and $d = 8$ have both been rounded to the nearest whole number.

- **a** Calculate the lower bound of the value of $c \times d$.
- **b** Calculate the upper bound of the value of $\frac{c}{d}$.

13 A school field has a $100\ \text{m}$ track that is measured to the nearest metre. The record for the fastest $100\ \text{m}$ sprint is 13.9 seconds rounded to the nearest 0.1 seconds.

- **a** What is the fastest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.
- **b** What is the slowest speed, in metres per second, the record holder could have run? Round your answer correct to two decimal places.

14 A circle has a radius of $7\ \text{cm}$, correct to the nearest centimetre.

- **a** Calculate its area correct to one decimal place.
- **b** What are the smallest and largest possible areas? Round your answer to one decimal place.
- **c** What is the maximum possible absolute error in the calculation?
- **d** What is the maximum possible percentage error in the calculation? Round your answer to one decimal place.

15 A field has dimensions $15.4\ \text{m} \times 17.6\ \text{m}$, to the nearest $10\ \text{cm}$.

- **a** What are the upper and lower bounds of the area of the field?
- **b** What is the upper and lower bounds of the perimeter of the field?

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16 A cube has a side length of $10\ \text{cm}$ rounded to the nearest centimetre.

a Find the lower and upper bounds of the side length.

b Find the area of one face of the cube, using the given side length.

c Find the lower and upper bounds of the area of one face.

d Find the volume of the cube using the given side length.

e Find the lower and upper bounds of the volume of the cube.

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